Problem 1

(Based on U&F 6th edition, problems 3.16, and 3.19) Consider a thin rectangular section cantilever beam of depth $2h$, length L, and thickness t, loaded by a uniform distributed load per unit length p (figure P3.19). You could estimate stresses in the beam using a Mechanics of Materials model, or using the more complex method of elasticity. In this problem I would like you to do both and compare your results. Set up your solution on a computer (e.g. in Excel, Matlab, or Python) so that you can easily change the dimensions. Turn in results for a structural steel beam, with baseline dimensions $h = 200$ mm, $t = 76$ mm, $L = 1.0$ m, $E = 200$ GPa, $\nu = 0.3$, and distributed load $p = 10$ kN/m.

Please save yourself and me a lot of trouble by using the coordinate system shown in the figure. (If you start the x axis at the base of the beam you will just add more terms to the Airy Stress Function that make the solution more complicated than necessary.)

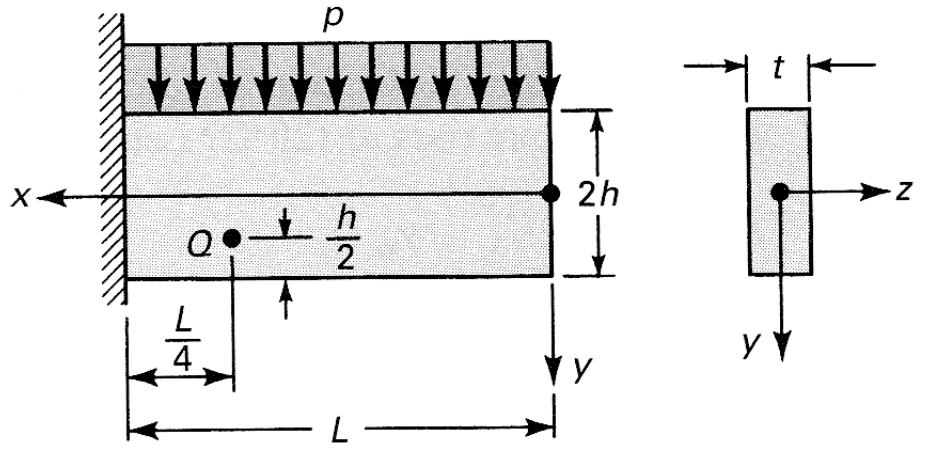


FIGURE P3.19.

Given:

- $h = 200\ mm$
- $t = 76\ mm$
- $L = 1\ m$
- $E = 200\ GPa$
- $\nu = 0.3$
- $p = 10\ kN/m$

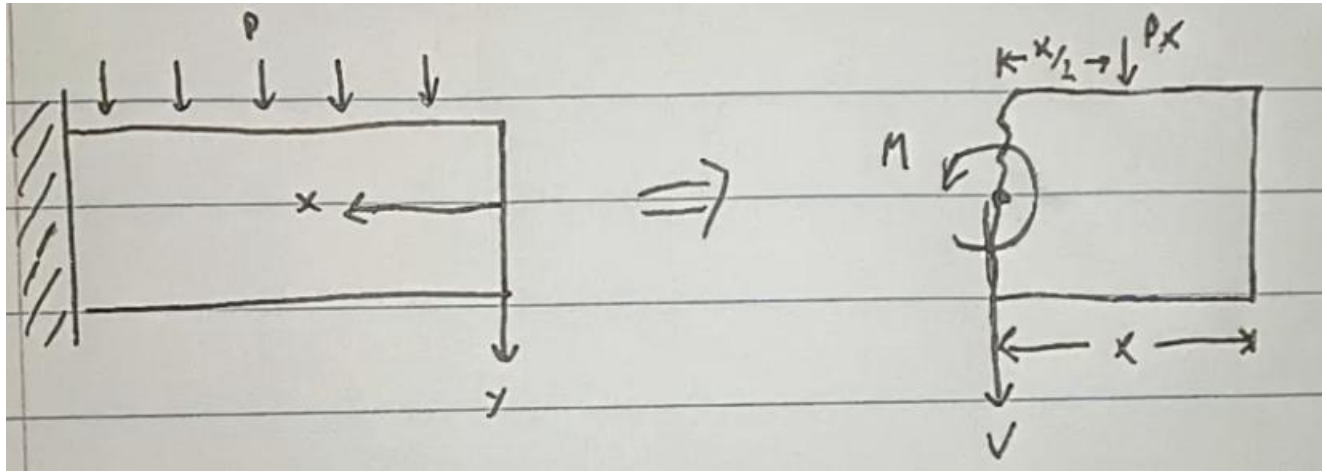
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Problem 1 – Part a

Mechanics of Materials Model: calculate and graph the internal shear force distribution $V(x)$ and bending moment distribution $M(x)$ in the beam. Then derive the expressions for normal stress and shear stress in the beam cross-section, $\sigma_x(x,y)$ and $\tau_{xy}(x,y)$. Plot the stress distributions $\sigma_x(y)$ and $\tau_{xy}(y)$ for a cross-section passing through point Q in the figure (located at $x = 3L/4$), plot axial stress distributions $\sigma_x(x)$ along the beam top surface ($y = -h$), and $\tau_{xy}(x)$ along the beam centerline ($y=0$), and report the stresses (with units) at point Q.

Approach:

- Starting from the free end of the beam (x=0), I will make a cut at an arbitrary position x and use static equilibrium to find the internal shear V(x) and moment M(x). Replacing the load distribution as a point load with its line of action through the centroid of the distribution shape. Noting the following sign convention for a +x face:



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Approach:

- Define geometry:

$$I = \frac{1}{12}(t)(2h)^3 = \frac{2th^3}{3}$$

  - Neutral axis pases through the centroid of the beam cross-section, thus it passes through $y = 0$
- The normal stress profile across a cross-section is given by the following:

$$\sigma_x(x, y) = -\frac{M_z(x)y}{I}$$

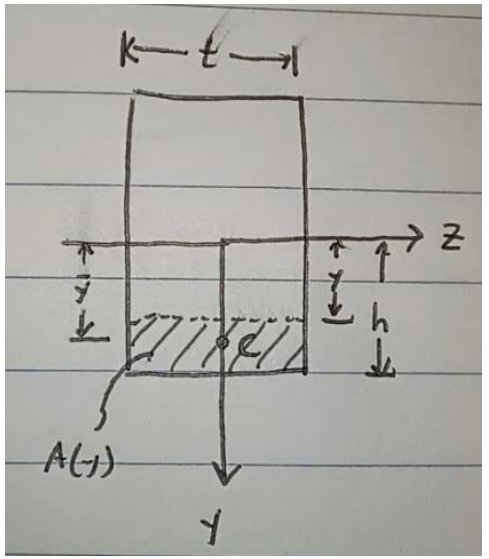
- The shear stress profile across a cross-section is given by the following:

$$\tau_{xy}(x, y) = \frac{V(x)Q(y)}{It}$$

  - Where the first area moment Q is found by:

$$Q(y) = \bar{y}A(y)$$

$$Q(y) = \left[y + \frac{h - y}{2}\right][(h - y)t]$$



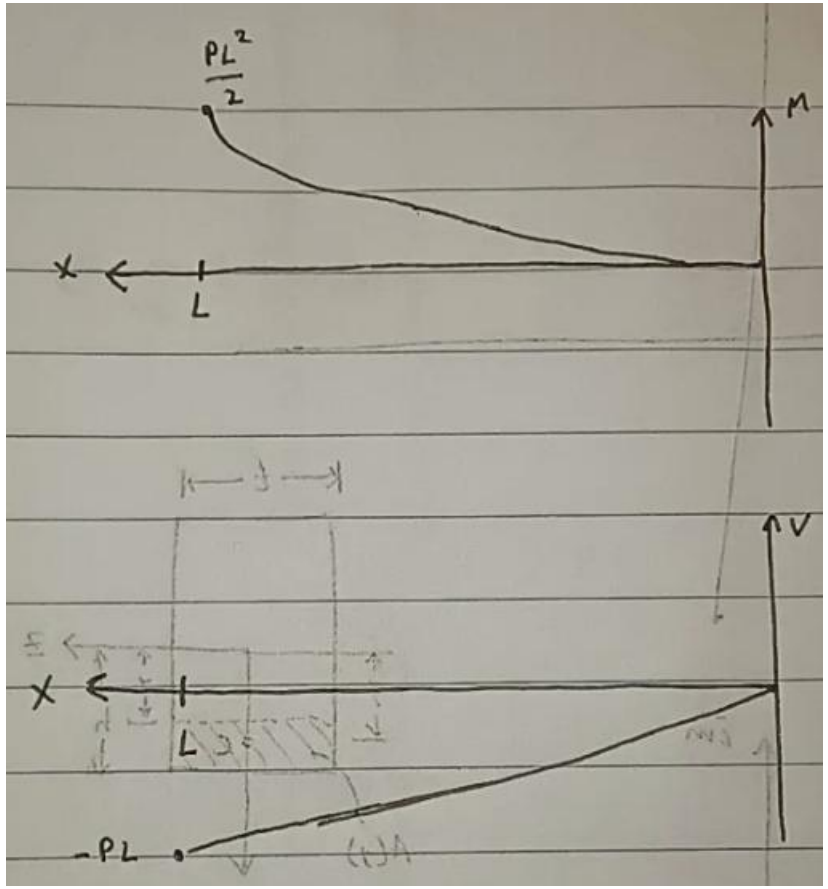
- Finally, I can plot the stress distributions using the derived equations on Python:
  - Normal stress σ_x at $x = 3L/4$ varying y
  - Shear stress τ_{xy} at $x = 3L/4$ varying y
  - Normal stress σ_x at $y = -h$ varying x
  - Shear stress τ_{xy} along the centerline/neutral axis $y = 0$ varying x
  - + report numerical stresses at point $Q = \left(\frac{3L}{4}, \frac{h}{2}\right)$

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Section FBD equilibrium:

$$\sum F_y = V + px = 0 \quad \rightarrow \quad \boxed{V(x) = -px}$$

$$\sum M_{z,\,x=0} = M_z + Vx + px\left(\frac{x}{2}\right) = 0 \quad \rightarrow \quad M_z(x) = -Vx - \frac{px^2}{2} \quad \rightarrow \quad \boxed{M_z(x) = \frac{px^2}{2}}$$



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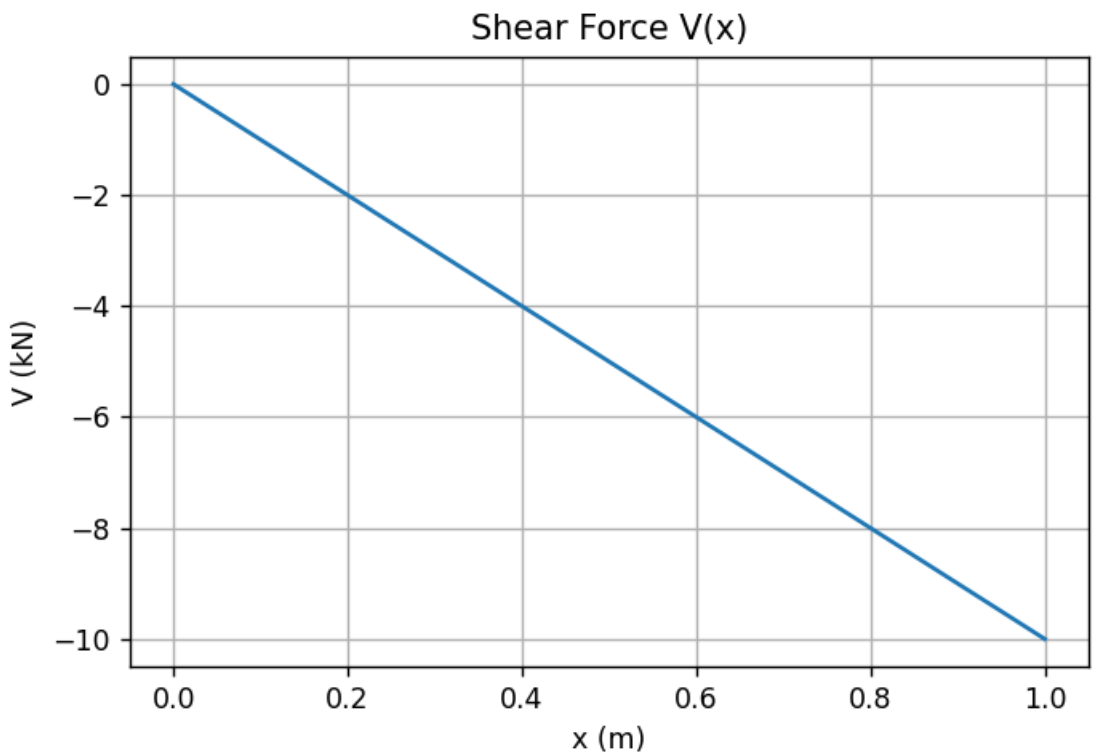
Solving for stress distributions symbolically:

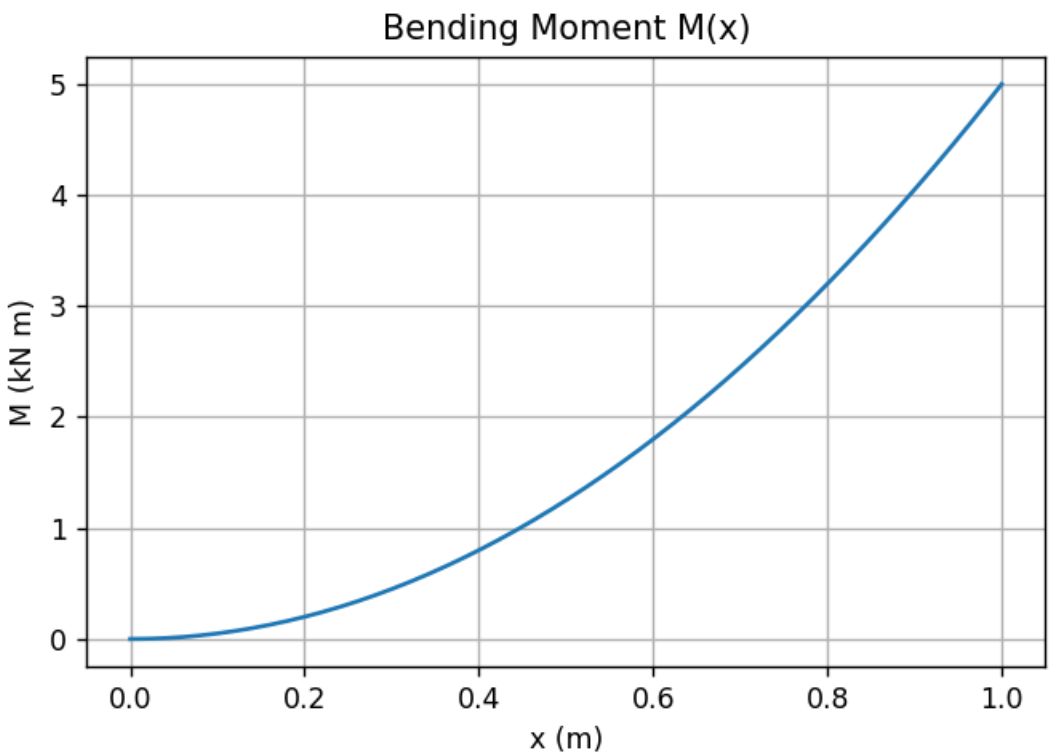
$$\sigma_x(x,y) = -\frac{M_z(x)y}{I} = -\frac{\frac{px^2}{2}}{\frac{2th^3}{3}}y \quad \rightarrow \quad \boxed{\sigma_x(x,y) = -\frac{3px^2y}{4th^3}}$$

$$Q(y) = \left[y + \frac{h-y}{2}\right][(h-y)t] = \frac{(h^2-y^2)t}{2}$$

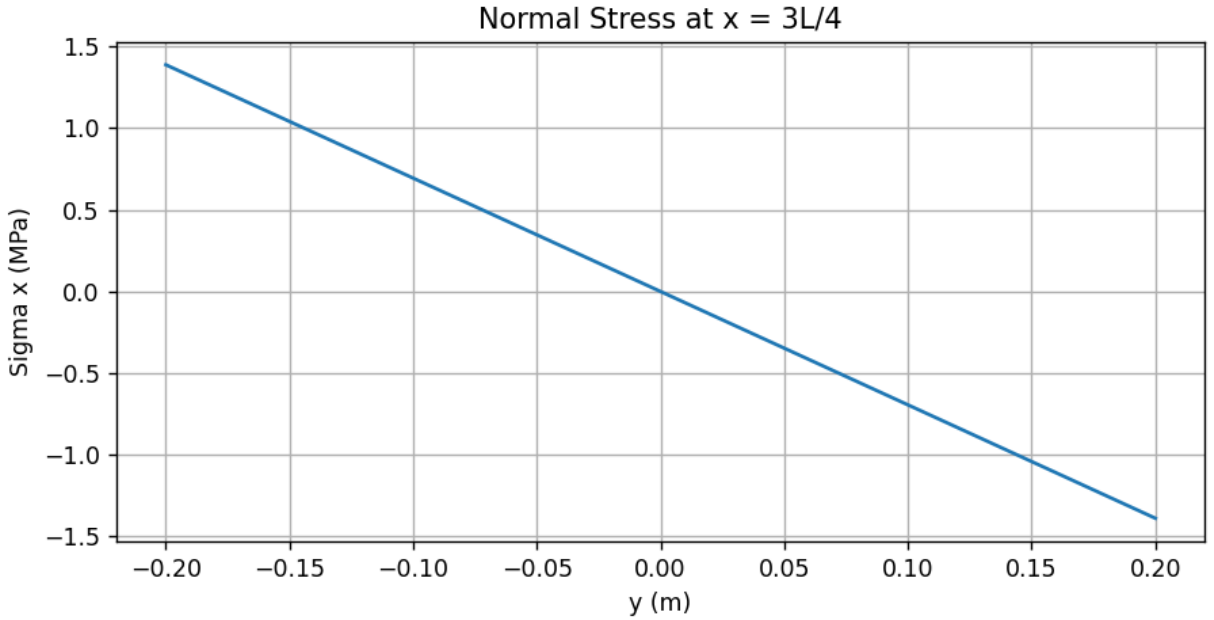
$$\tau_{xy}(x,y) = \frac{V(x)Q(y)}{It} = \frac{-px\frac{(h^2-y^2)t}{2}}{\frac{2th^3}{3}t} \quad \rightarrow \quad \boxed{\tau_{xy}(x,y) = \frac{-3px(h^2-y^2)}{4th^3}}$$

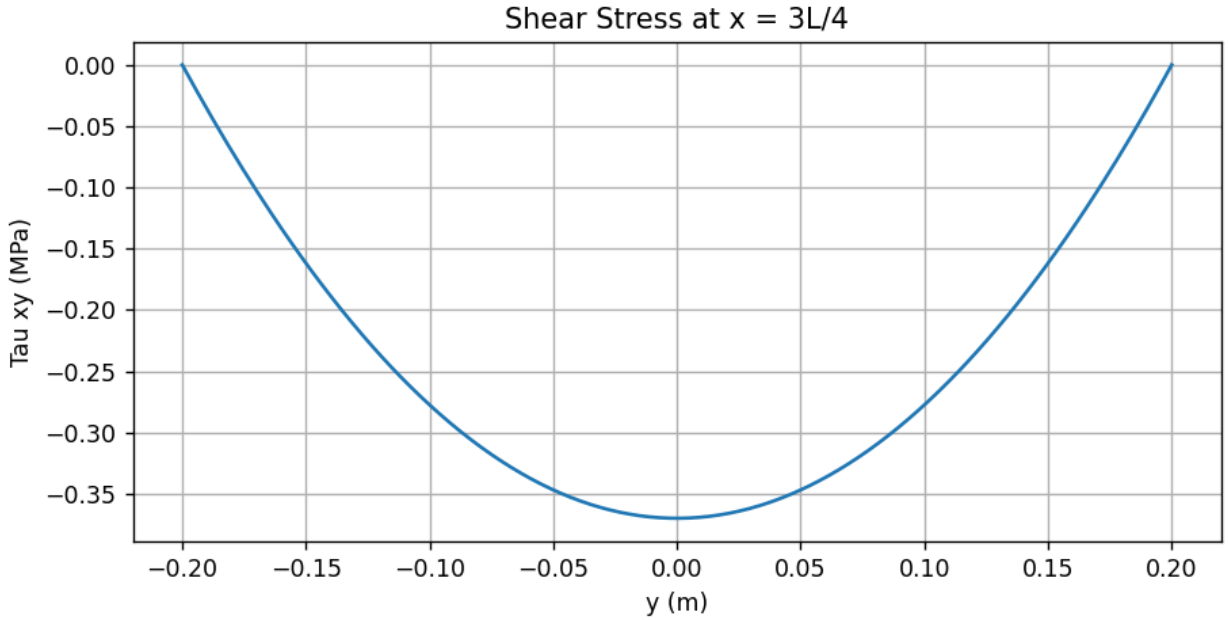
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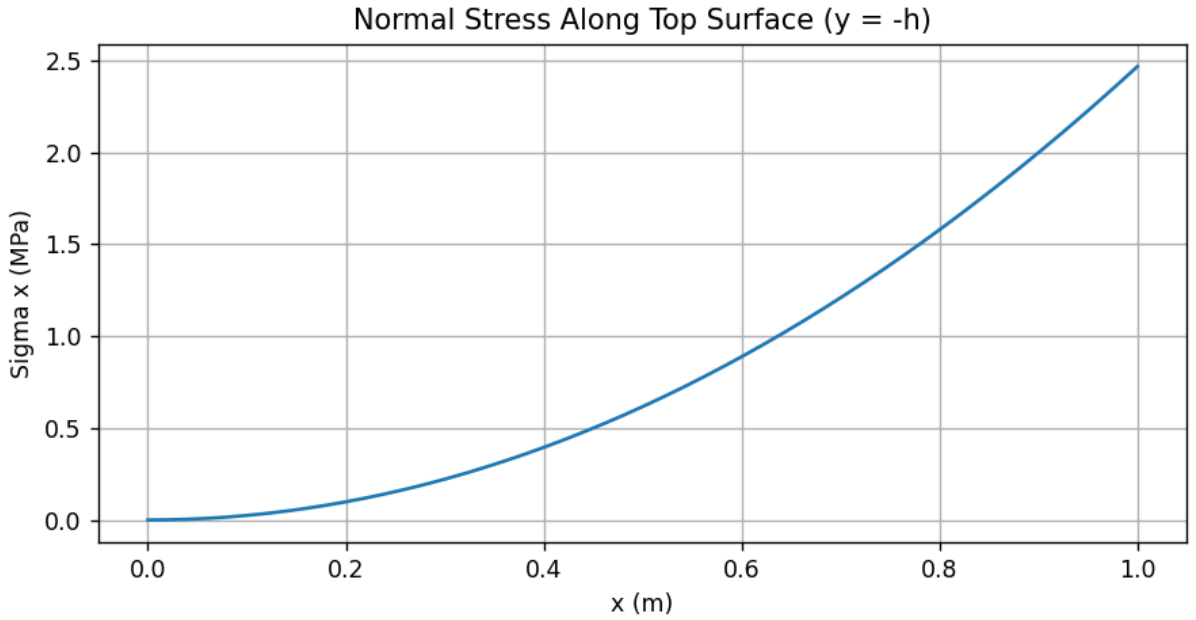


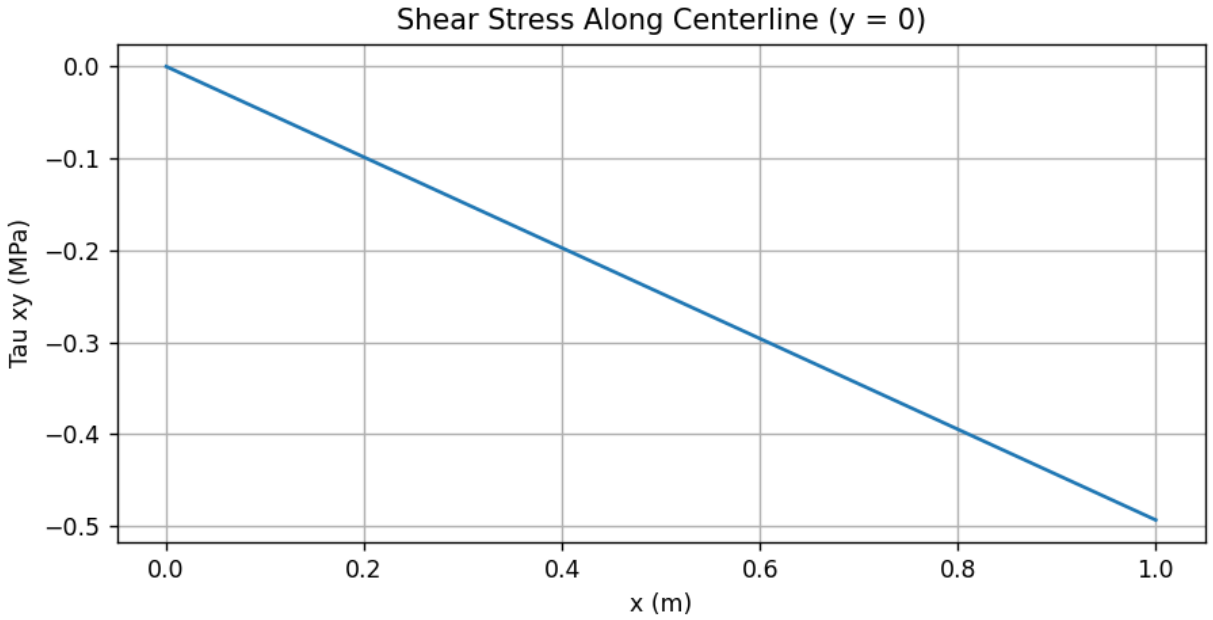


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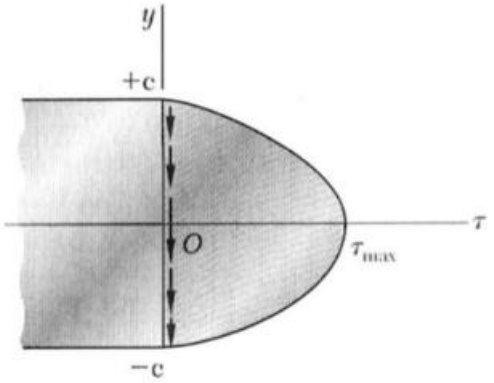


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Stresses at point Q

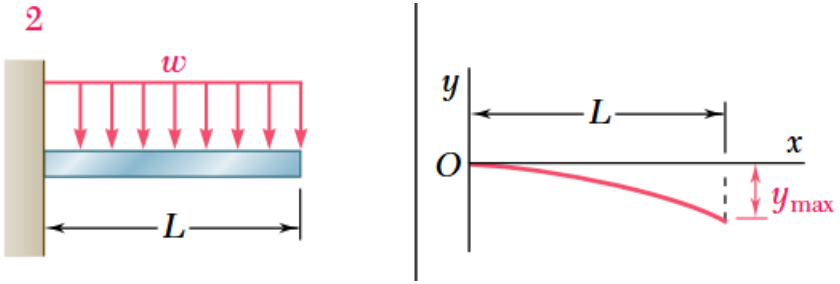
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Point Q: x = 0.750 m, y = 0.100 m
Sigma x at Q = -0.6939 MPa
Tau xy at Q = -0.2775 MPa
```



These results make sense because the boundary conditions are met:

- Zero shear stress at the top and bottom surfaces
- Max shear stress at the neutral axis (y=0, centerline)
- Zero normal stress at neutral axis
- Zero shear and normal stress at the free end x=0

Additionally, the signs for the stresses at point Q makes sense. Since the normal stress would be compressive (-) and the shear stress would be pointing along the negative y direction, matching the direction of the internal shear force V. [Time spent: 1hr]



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Problem 1 – Part b

Method of Elasticity Model: Consider the following Airy stress function (ASF) to model beam stress and deformation:

$$\Phi = ax^2 + bx^2y + cy^3 + dy^5 + ex^2y^3$$

where coefficients *a* through *e* are constants. Discuss briefly why this ASF makes sense, based on what we saw modeling a point-loaded beam carrying a uniform internal shear force, and/or based on the MechMat model in part a. (i) Determine conditions that must be satisfied so that Φ is biharmonic. (ii) Then determine the stress fields, $\sigma_x(x,y)$, $\sigma_y(x,y)$, and $\tau_{xy}(x,y)$. (Hint: Use the unloaded bottom and compression loaded top to define boundary conditions on the stress fields, and use an integral of stress over a cross-section at location *x* to define the unknown constants in Φ.) (**Note:** don't use a traction-free boundary condition at the tip of the beam; if you do you won't find any solution.)

Discussion:

- The airy stress function is a scalar function from which stress components are defined to automatically satisfy equilibrium and results in the following stress fields for plane stress:

$$\sigma_x = \frac{\partial^2\Phi}{\partial y^2}, \qquad \sigma_y = \frac{\partial^2\Phi}{\partial x^2}, \qquad \tau_{xy} = -\frac{\partial^2\Phi}{\partial x \partial y}$$

  - The airy function, like any stress field, must also satisfy strain compatibility through the biharmonic equation (homogenous, isotropic, linear elastic materials with negligible body forces). Solving the biharmonic would give us relationships/constraints on the coefficients of the airy stress function, thus limiting the possible functions.

$$\nabla^4\Phi = \nabla^2(\nabla^2\Phi) = 0 \quad \text{(Biharmonic Eqn)}$$

$$\frac{\partial^4\Phi}{\partial x^4} + 2\frac{\partial^4\Phi}{\partial x^2\partial y^2} + \frac{\partial^4\Phi}{\partial y^4} = 0$$

  - Finally, we would use boundary conditions and known resultant loads to fully determine the airy stress function coefficients for our particular problem set up.

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Discussion:

- Why is the airy stress function ($\Phi = ax^2 + bx^2y + cy^3 + dy^5 + ex^2y^3$) make sense?
  - Looking at its corresponding stress fields:

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2} = 6cy + 20dy^3 + 6ex^2y$$

$$\sigma_y = \frac{\partial^2 \Phi}{\partial x^2} = 2a + 2by + 2ey^3$$

$$\tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y} = -2bx - 6exy^2$$

  - Compare functional forms with the stress distributions found in part a (mech mat model). Note that normal stress in the y direction was not considered for the Mech Mat model.

Stress	Mech Mat Model	Airy stress function
σ_x	$\frac{3px^2y}{4th^3}$	$\propto x^2y$
τ_{xy}	$\frac{-3px(h^2 - y^2)}{4th^3}$	$\propto x, xy^2$

Noting that while the general forms of the stress fields derived from the Airy stress function match those of the Mech Mat model, there are additional terms that may contribute to more complex stress distributions than the simplified Mech Mat solution.

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Approach:

- Determine the constraints on airy stress function coefficients such that it satisfies compatibility (biharmonic equation)

$$\frac{\partial^4 \Phi}{\partial x^4} + 2\frac{\partial^4 \Phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \Phi}{\partial y^4} = 0$$

$$24ey + 120dy = 0$$

- Apply boundary conditions:
  - Zero traction on lower surface y=h

$$\sigma_y(h) = 0, \qquad \tau_{xy}(h) = 0$$

  - Only load on upper surface y=-h is p

$$\sigma_y(-h) = -\frac{p}{t}, \qquad \tau_{xy}(-h) = 0$$

  Note that p is (force)/(length in x) so dividing again by length in z gives pressure.
- Apply known resultant internal loads. Picking the free end since its zero moment and shear:
  - Axial force $P_x(0) = 0$

$$P_x = \int \sigma_x \, dA = t \int_{-h}^{h} \sigma_x dy$$

$$P_x = t \int_{-h}^{h} 6cy + 20dy^3 + 6ex^2 y \, dy = 0$$

  The integral of an odd function wrt y over a symmetric interval is zero.

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Approach:

- Shear force $V(0) = 0$

$$V = \int \tau_{xy}\, dA = t \int_{-h}^{h} \tau_{xy} dy$$

$$V = t \int_{-h}^{h} -2bx - 6exy^2\, dy$$

$$V(0) = 0$$

Not very useful

- Moment $M(0) = 0$

$$M = \int -\sigma_x y\, dA = -t \int_{-h}^{h} 6cy^2 + 20dy^4 + 6ex^2y^2\, dy$$

$$M(0) = -t \int_{-h}^{h} 6cy^2 + 20dy^4\, dy = -t[2cy^3 + 4dy^5]\,{}_{-h}^{h} = -t(4ch^3 + 8dh^5) = 0$$

Useful. Note that for pos. tensile stress in the lower half of the beam and pos. y down, $r \times F$ gives a moment into the page which is in the $-z$ direction.

Note: While we know that the resultant bending moment and shear force at the free end are zero, we cannot directly say that $\sigma_x(0, y) = 0$ for all y. This condition applies to every point and is too restrictive for our assumed Airy stress function, since positive and negative stresses can combine to produce a net zero resultant. While the actual free end should be traction-free at every point, our assumed polynomial cannot satisfy this condition along with the other boundary conditions. Therefore, we consider the resultant forces and moments by integrating the stress distributions over the cross-section instead of directly defining the stresses at every point.

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Constraints on Coef from Biharmonic

$$24ey + 120dy = 0 \longrightarrow \boxed{5d + e = 0}$$

Constraints on Coef from Boundary Conditions

$$\sigma_y(h) = 0 \longrightarrow 2a + 2bh + 2eh^3 = 0 \longrightarrow \boxed{a + bh + eh^3 = 0}$$

$$\tau_{xy}(h) = 0 \longrightarrow -2bx - 6exh^2 = 0 \longrightarrow \boxed{b + 3eh^2 = 0}$$

$$\sigma_y(-h) = -\frac{p}{t} \longrightarrow 2a - 2bh - 2eh^3 = -\frac{p}{t} \longrightarrow \boxed{a - bh - eh^3 = -\frac{p}{2t}}$$

$$\tau_{xy}(-h) = 0 \longrightarrow -2bx - 6exh^2 = 0 \longrightarrow b + 3eh^2 = 0$$

Now we have 4 indep eqns, 5 unknowns. Get last eqn from moment resultant:

$$M(0) = -t(4ch^3 + 8dh^5) = 0 \longrightarrow \boxed{c + 2dh^2 = 0}$$

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Solve system of eqns for coefficients:

1. $5d + e = 0$
2. $a + bh + eh^3 = 0$
3. $b + 3eh^2 = 0$
4. $a - bh - eh^3 = -\frac{p}{2t}$
5. $c + 2dh^2 = 0$

$(eqn\ 2) + (eqn\ 4)$:

$$2a = -\frac{p}{2t} \longrightarrow \boxed{a = -\frac{p}{4t}}$$

$\frac{h}{3}(eqn\ 3) + (eqn\ 4)$:

$$\left(\frac{1}{3}bh + eh^3\right) + (a - bh - eh^3) = -\frac{p}{2t} \longrightarrow a - \frac{2}{3}bh = -\frac{p}{2t} \longrightarrow -\frac{2}{3}bh = -\frac{p}{2t} + \frac{p}{4t} \longrightarrow \boxed{b = \frac{3p}{8th}}$$

$(eqn\ 3)$:

$$b + 3eh^2 = 0 \longrightarrow 3eh^2 = -\frac{3p}{8th} \longrightarrow \boxed{e = -\frac{p}{8th^3}}$$

$(eqn\ 1)$:

$$5d + e = 0 \longrightarrow 5d = \frac{p}{8th^3} \longrightarrow \boxed{d = \frac{p}{40th^3}}$$

$(eqn\ 5)$:

$$c + 2dh^2 = 0 \longrightarrow \boxed{c = -\frac{p}{20th}}$$

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Resulting Airy Stress Function and Stress Fields:

$$\boxed{\Phi = -\frac{p}{4t}x^2 + \frac{3p}{8th}x^2y - \frac{p}{20th}y^3 + \frac{p}{40th^3}y^5 - \frac{p}{8th^3}x^2y^3}$$

$$\boxed{\sigma_x(x,y) = \frac{\partial^2\Phi}{\partial y^2} = -\frac{3p}{10th}y + \frac{p}{2th^3}y^3 - \frac{3p}{4th^3}x^2y}$$

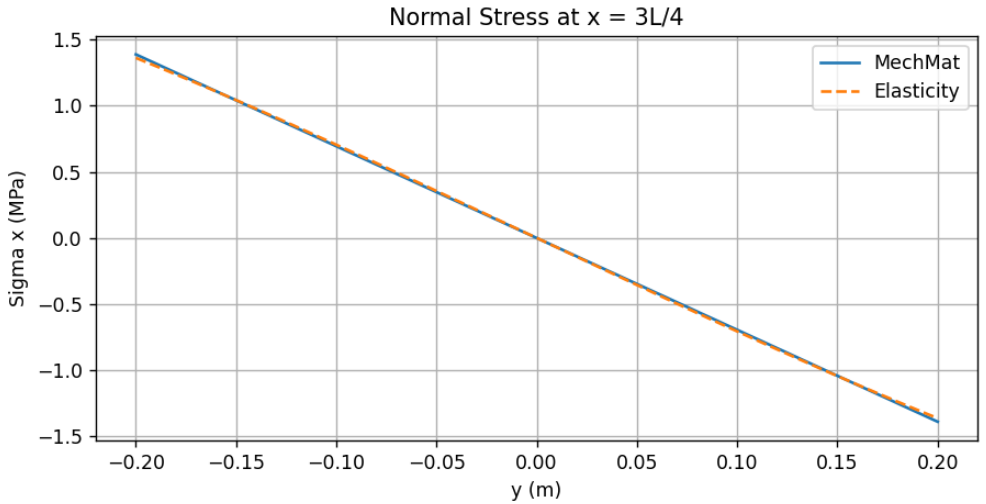
$$\boxed{\sigma_y(x,y) = \frac{\partial^2\Phi}{\partial x^2} = -\frac{p}{2t} + \frac{3p}{4th}y - \frac{p}{4th^3}y^3}$$

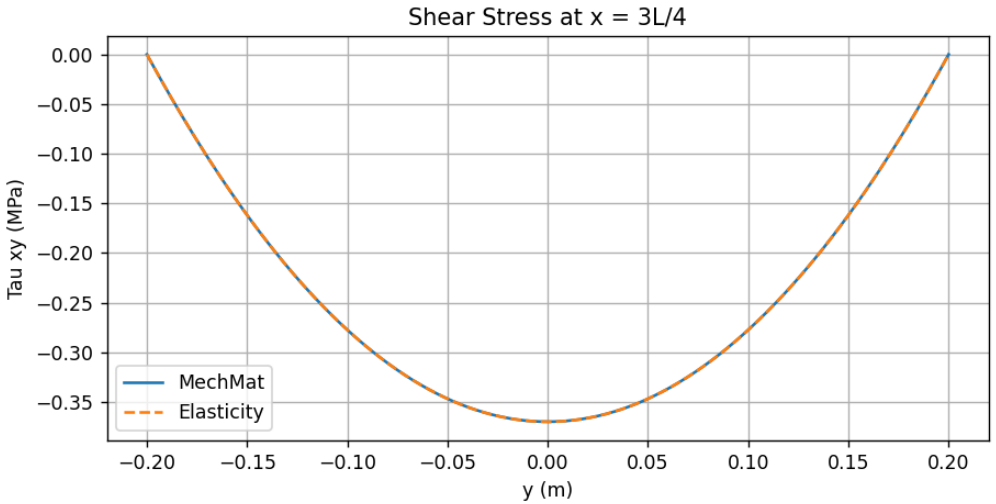
$$\boxed{\tau_{xy} = -\frac{\partial^2\Phi}{\partial x\partial y} = -\frac{3p}{4th}x + \frac{3p}{4th^3}xy^2}$$

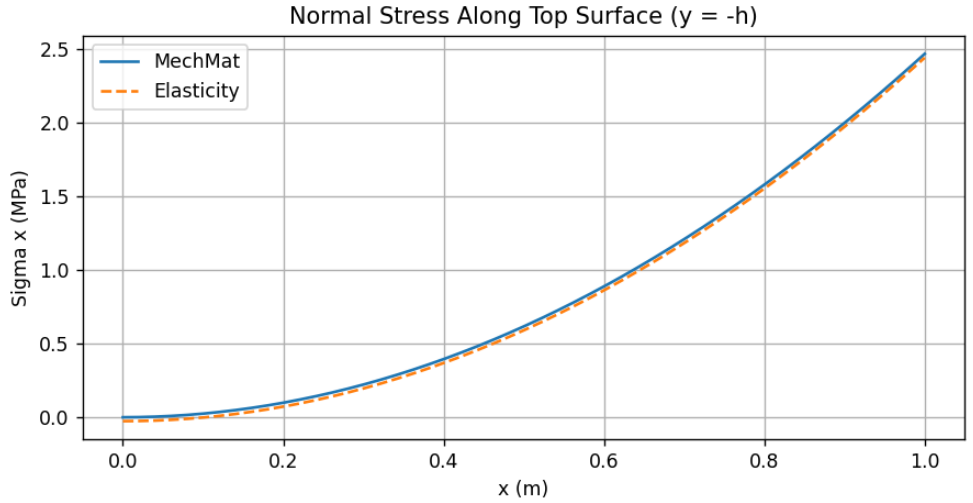
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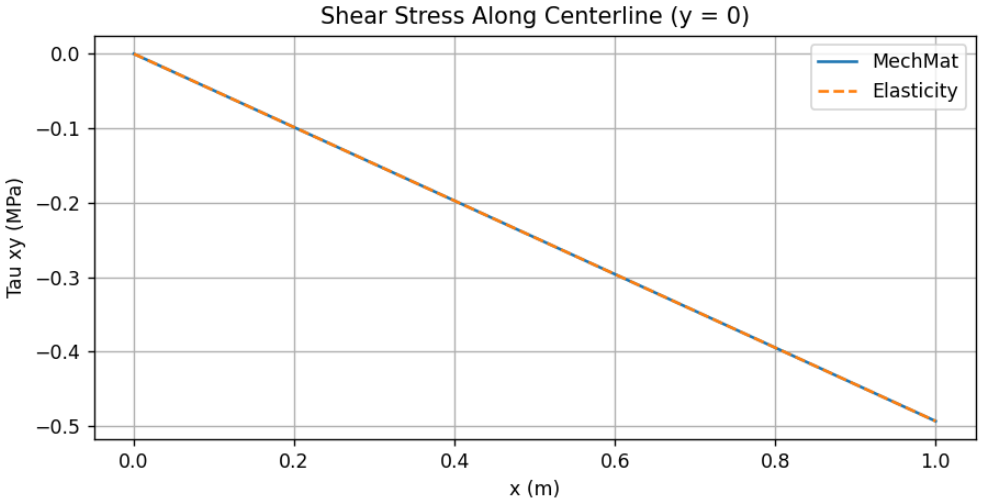
Problem 1 – Part c

Compare the elasticity and MechMat solutions by including the method of elasticity stresses in the plots of stresses in the cross-section and axial stress distributions, as well as comparing the stresses computed at point *Q*.









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Part C: Stress Comparison at Q (MPa)
MechMat:    Sigma x = -0.6939, Tau xy = -0.2775
Elasticity: Sigma x = -0.7054, Sigma y = -0.0206, Tau xy = -0.2775
```

Problem 1 – Part d

Which model is more accurate, and why? Where are the differences greatest?

Both models generally agree with each other, but there are small differences, particularly in the normal stress distributions. It appears that the MechMat model predicts slightly higher axial stress magnitudes in some regions compared to the method of elasticity. Meanwhile the shear stress distribution matches exactly between the two methods. [Time spent: 45 mins]

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Problem 1 – Part e

Finally, vary the beam length and look at how your plots change. At what length to depth ratio is the maximum difference between the two methods over 10%?

Approach:

- Define length to depth ratio

$$\lambda = \frac{L}{2h}$$

A smaller ratio represents a shorter, wider beam, while a larger ratio represents a longer, more slender beam

- Find the difference of the two normal stress distributions:
  - Mech mat: $\sigma_x(x,y) = -\frac{3px^2y}{4th^3}$
  - Elasticity: $\sigma_x(x,y) = \frac{\partial^2\Phi}{\partial y^2} = -\frac{3p}{10th}y + \frac{p}{2th^3}y^3 - \frac{3p}{4th^3}x^2y$

$$\Delta\sigma_x = -\frac{3p}{10th}y + \frac{p}{2th^3}y^3$$

  This is not dependent on L.

- Find the difference of the two shear stress distributions:
  - Mech mat: $\tau_{xy}(x,y) = \frac{-3px(h^2-y^2)}{4th^3}$
  - Elasticity: $\tau_{xy} = -\frac{\partial^2\Phi}{\partial x\partial y} = -\frac{3p}{4th}x + \frac{3p}{4th^3}xy^2$

$$\Delta\tau_{xy} = 0$$

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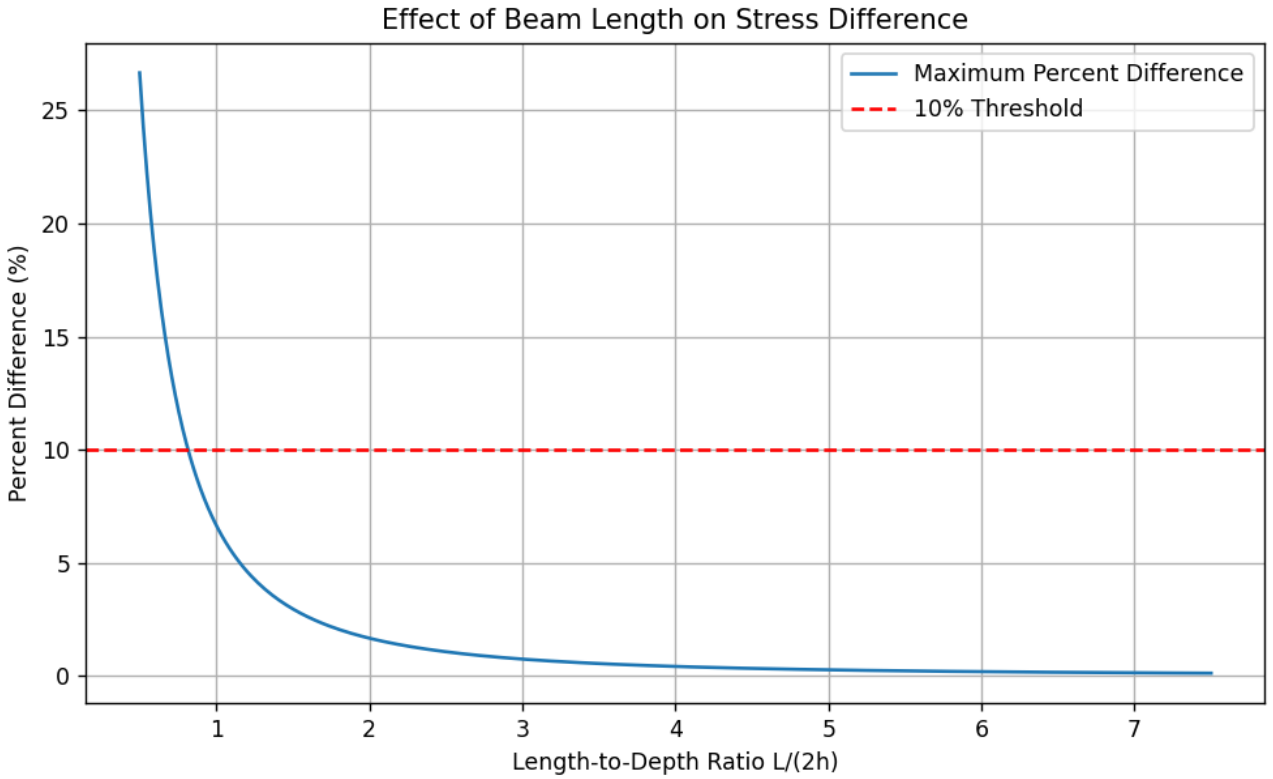
Approach:

- Define percent difference to see when it exceeds 10%, relative to mech mat axial stress:

$$\%\ Difference = \frac{max|\Delta\sigma_x|}{max|\sigma_{x,mechmat}|} \times 100$$

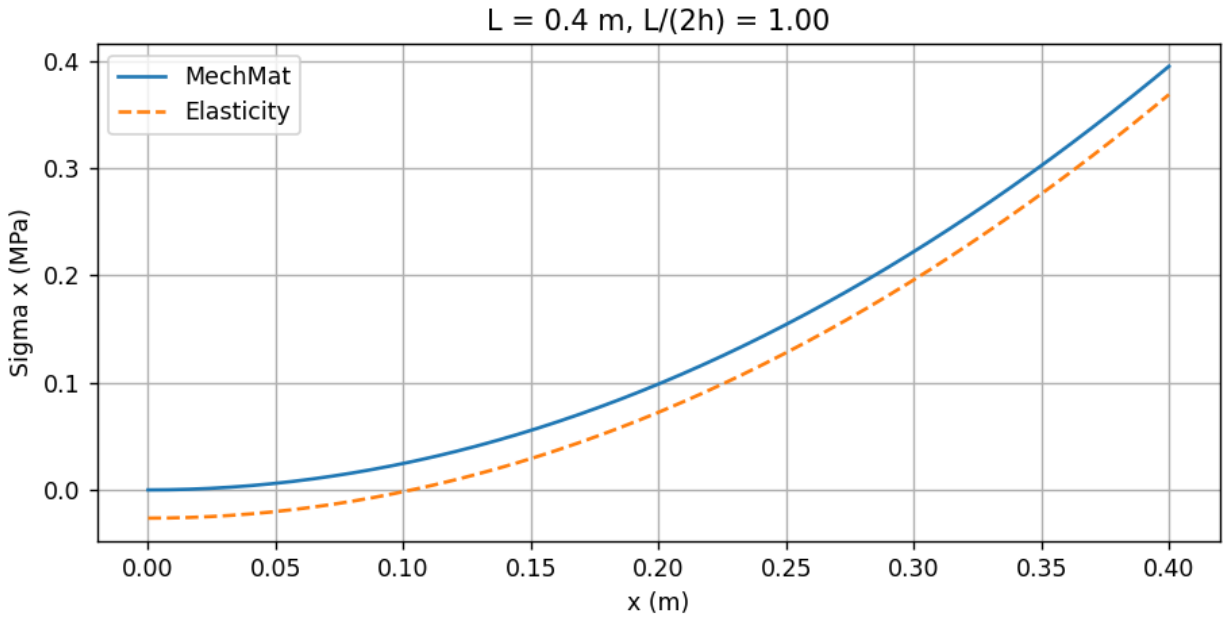
Using Python, keeping h,t,p constant and varying length L, I can calculate the stress fields for each length L. Then, using the maximum values of the difference in stresses and mech mat stress, I can calculate the max percent difference for each L.

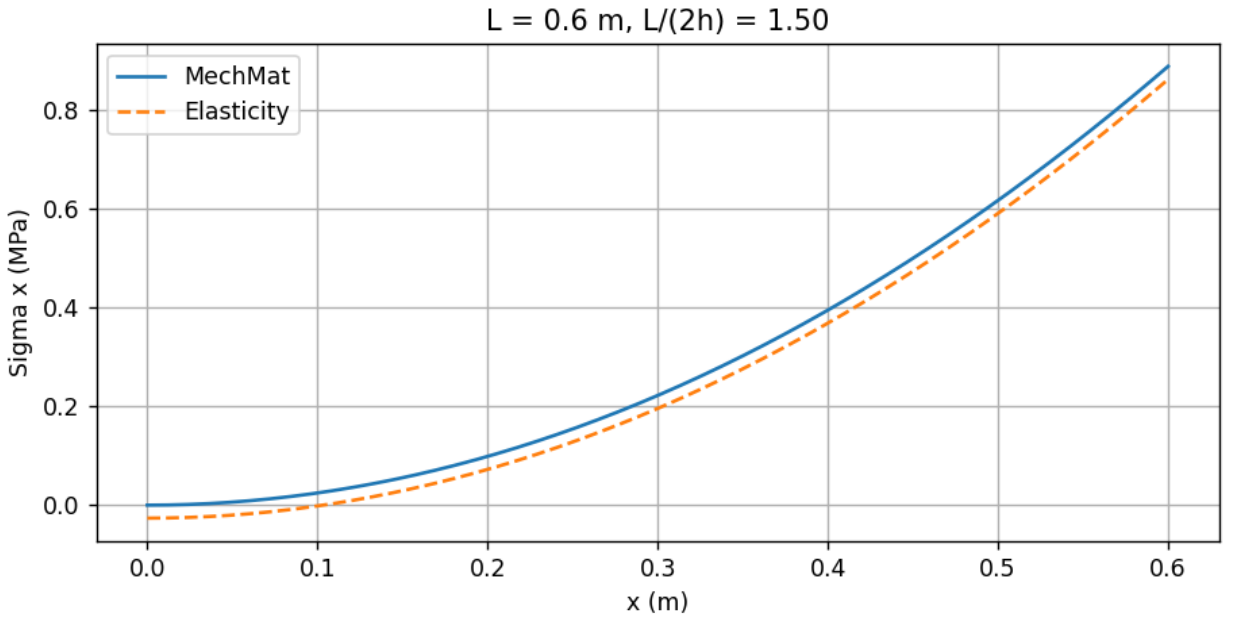
Finally, comparing the percent difference against L/2h would tell us when it hits the threshold.

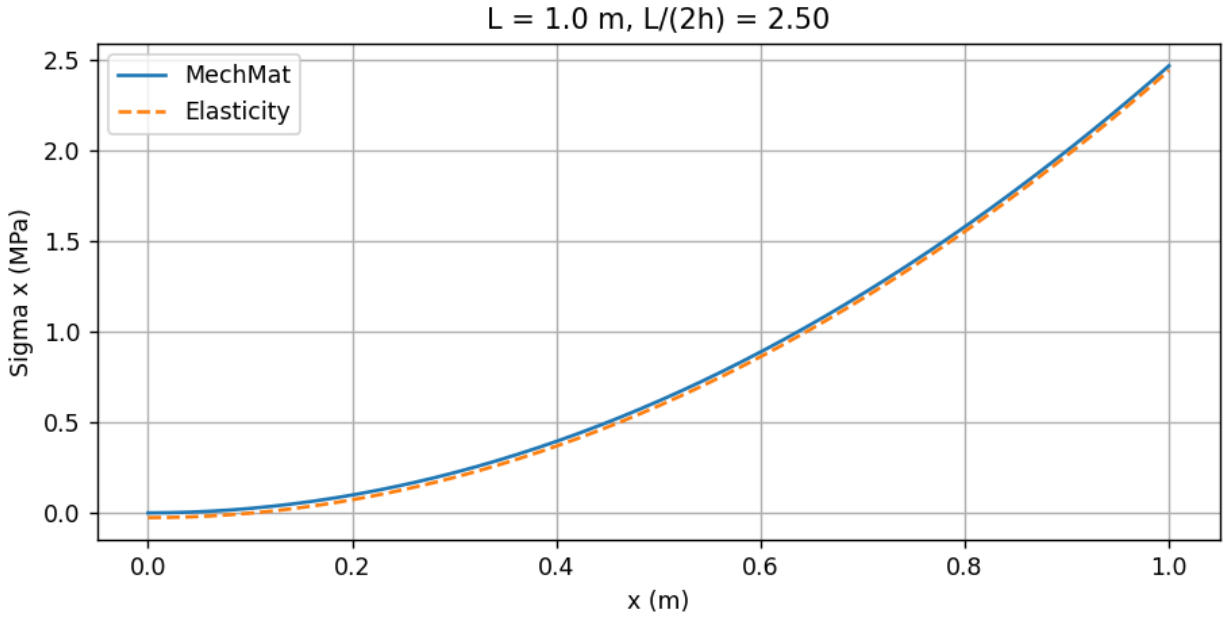


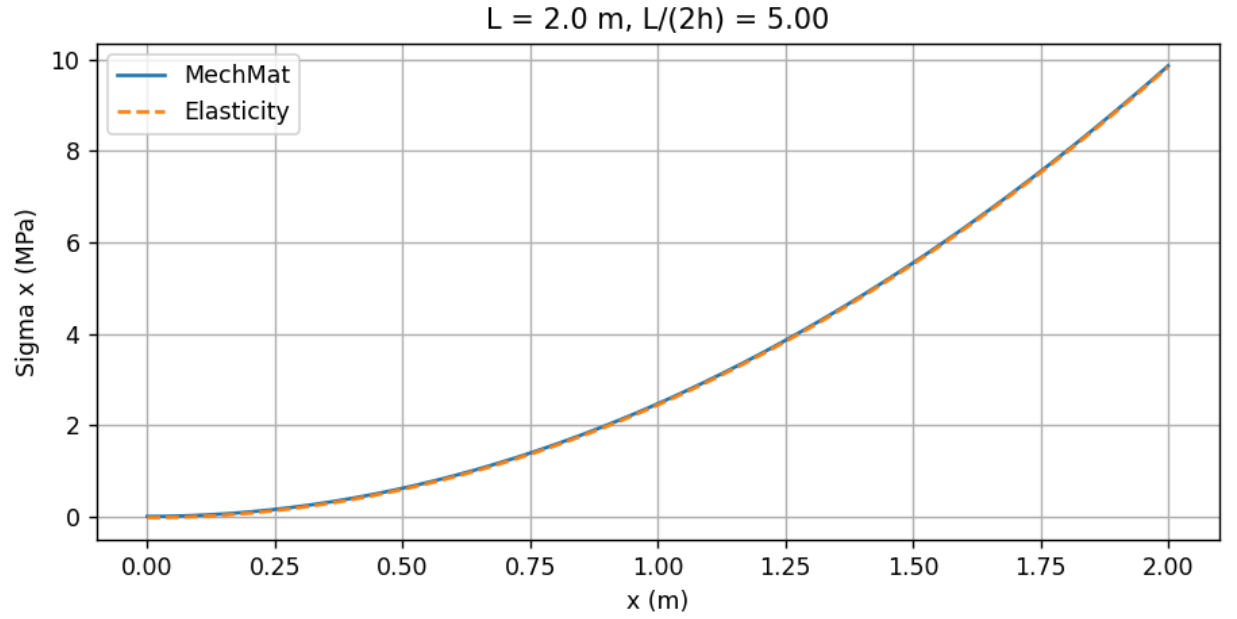
It exceeds the threshold when L = 0.327 m or a length to depth ratio of L/(2h)=0.817

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Problem 2

Give a brief overview of how the Airy Stress Function works for solving 2D method of elasticity problems. How does the Airy Stress Function guarantee stress equilibrium in plane stress or plane strain problems when body forces are negligible? How is compatibility enforced? It may help to refer to parts of problem 1 to explain.

Force equilibrium with zero body forces gives:

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0, \qquad \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} = 0$$

Strain compatibility combined with hooke's law for plane stress gives:

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \quad (compatibility)$$

$$\varepsilon_x = \frac{\sigma_x - \nu \sigma_y}{E}, \qquad \varepsilon_y = \frac{\sigma_y - \nu \sigma_x}{E}, \qquad \gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\nu)}{E} \tau_{xy} \quad (hooke's\ law)$$

Define airy stress function now:

$$\sigma_x = \frac{\partial^2 \Phi}{\partial y^2}, \qquad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2}, \qquad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y}$$

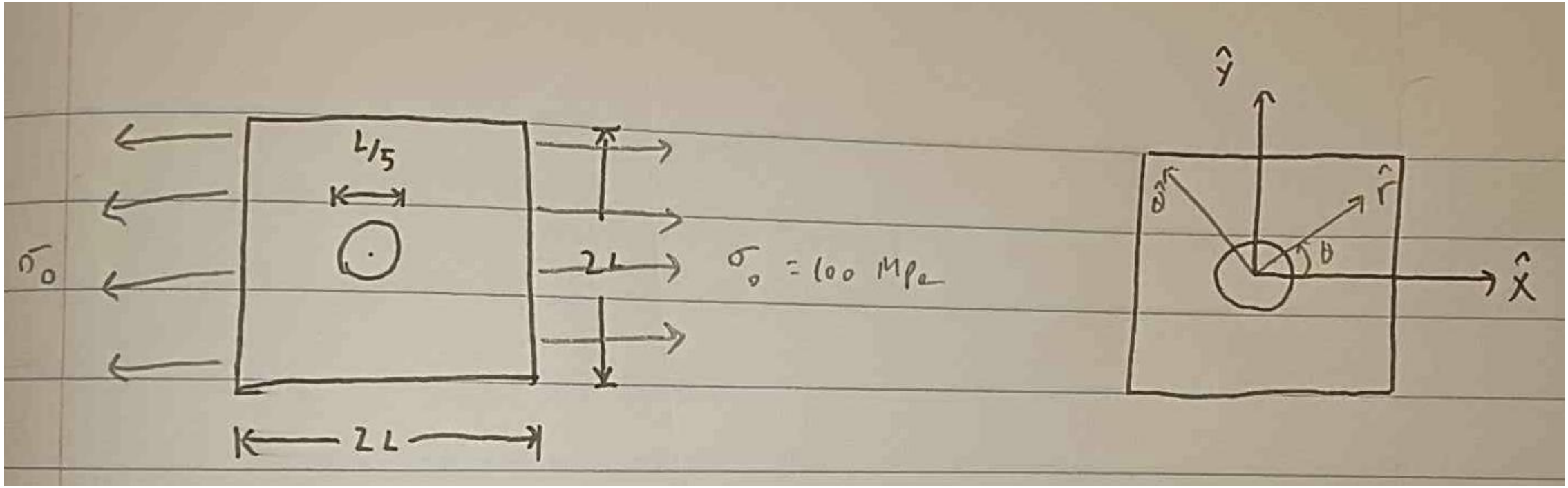
Plugging new definition into compatibility gives the biharmonic equation.

By definition, the airy stress function is built off of the stress equilibrium condition, thus ANY airy stress function will satisfy equilibrium. Additionally, compatibility is enforced through the biharmonic equation, since substituting the Airy stress function relations into the stress compatibility equation reduces it to the biharmonic equation.

Problem 3

Use Matlab or Python to create contour plots of the stress field in a square plate with a circular hole in the center, loaded with uniform stress, $\sigma_x(y) = \sigma_0 = 100$ MPa at the ends ($x = \pm L$). Make the hole radius 1/5 of the plate half-width. What is the stress concentration factor for this model? Make 10 stress contour plots, with legends or contour labels and indicating units, showing the following 10 stresses:

a. Normal stresses in cartesian and polar coordinates, σ_x, σ_y, σ_r and σ_θ
b. Shear stresses τ_{xy} and $\tau_{r\theta}$
c. Maximum and minimum principal stresses in the xy plane
d. Maximum shear stress (considering any orientation), and Von Mises stress



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Approach - Part a+b:

- Stresses in polar coordinates are given by superposition of a uniform axial stress field and a correction stress field that cancels the traction along the inner surface of the hole of radius a such that $\sigma_{rr}(a,\theta),\ \ \tau_{r\theta}(a,\theta)=0$.
  - Additional boundary conditions are:
    - $\sigma_{xx}(\pm L,y)=\sigma_0,\ \ \tau_{xy}(\pm L,y)=0$ (along the axial faces)
    - $\sigma_{yy}(x,\pm L)=0,\ \ \tau_{xy}(x,\pm L)=0$ (along the top and bottom)

This approximation is valid when the hole radius is much smaller than the length of the plate.

$$\sigma_{rr}=\frac{1}{2}\sigma_0\left[\left(1-\frac{a^2}{r^2}\right)+\left(1-\frac{4a^2}{r^2}+\frac{3a^4}{r^4}\right)\cos 2\theta\right]$$

$$\sigma_{\theta\theta}=\frac{1}{2}\sigma_0\left[\left(1+\frac{a^2}{r^2}\right)-\left(1+\frac{3a^4}{r^4}\right)\cos 2\theta\right]$$

$$\tau_{r\theta}=\frac{1}{2}\sigma_0\left(-1-\frac{2a^2}{r^2}+\frac{3a^4}{r^4}\right)\sin 2\theta$$

(see Ugural and Fenster §3.12.1, equation 3.55, but note there are errors in the 6th edition)

- To find the corresponding stresses in cartesian, we can use stress transformation equations:

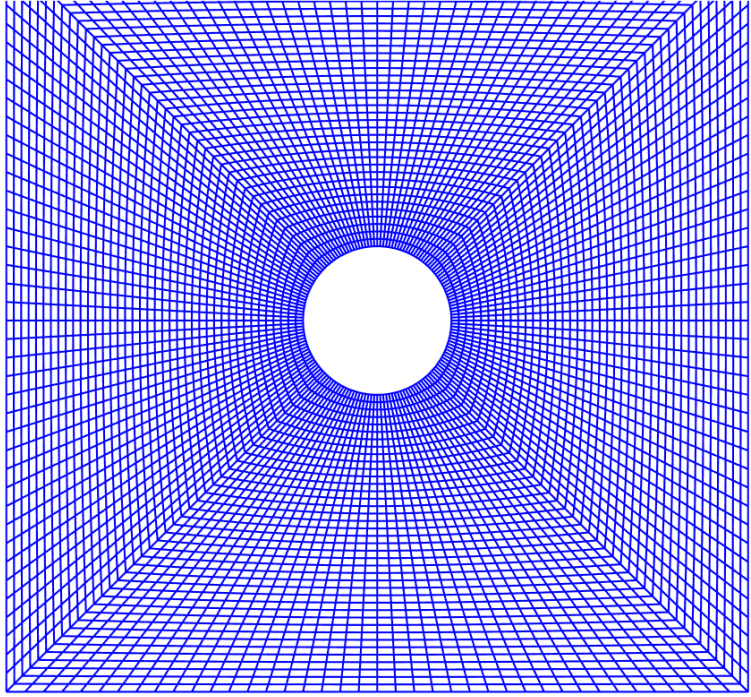
$$\sigma_{x'}=\sigma_{ave}+\frac{\Delta\sigma}{2}\cos 2\theta+\tau_{xy}\sin 2\theta$$

$$\sigma_{y'}=\sigma_{ave}-\frac{\Delta\sigma}{2}\cos 2\theta-\tau_{xy}\sin 2\theta$$

$$\tau_{x'y'}=\tau_{xy}\cos 2\theta-\frac{\Delta\sigma}{2}\sin 2\theta$$

Where $\sigma_{ave}=\frac{\sigma_x+\sigma_y}{2},\ \ \Delta\sigma=\sigma_x-\sigma_y$. Note that for the transformation from $r\theta$ to xy, the angle of transformation is $-\theta$.

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Approach – part c+d:

For the following plate mesh, I can do the following calculations at every point (x_{ij}, y_{ij}). Where i corresponds to angular subdivisions and j corresponds to radial subdivisions:

- Find the stresses $\sigma_x(x_{ij}, y_{ij}), \sigma_y(x_{ij}, y_{ij}), \tau_{xy}(x_{ij}, y_{ij})$
- Find

$$\sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}, \qquad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

- Max and min principal stresses are:

$$\sigma_1 = \sigma_{ave} + R, \qquad \sigma_2 = \sigma_{ave} - R$$

Since the out-of-plane shear stresses are zero, it is a principal direction with $\sigma_3 = 0$ (plane stress).

- Max shear stress for any orientation, including out of plane (3d) state of stress, is:

$$\tau_{max} = \frac{\max(\sigma_1, \sigma_2, \sigma_3) - \min(\sigma_1, \sigma_2, \sigma_3)}{2}$$

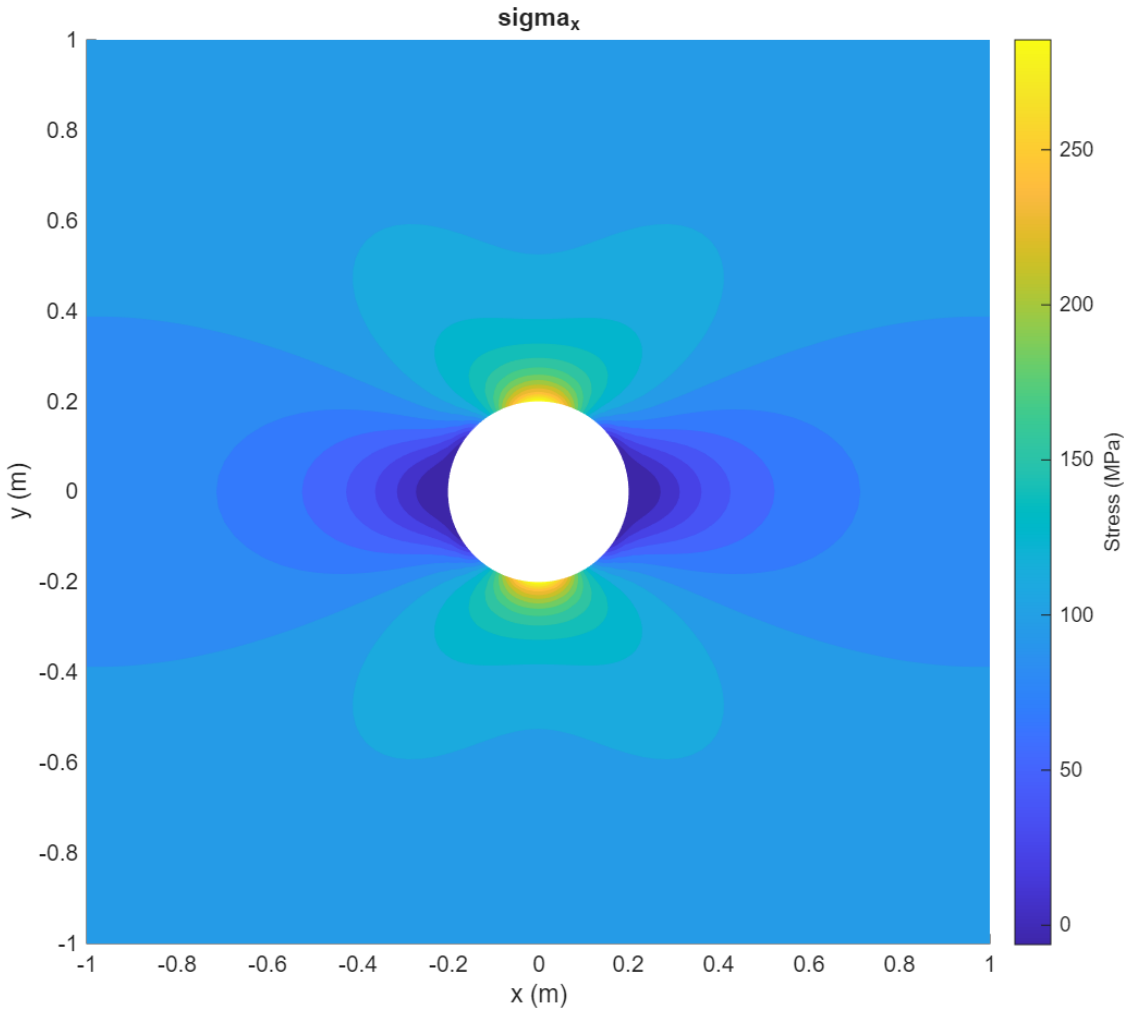
- Von-mises stress:

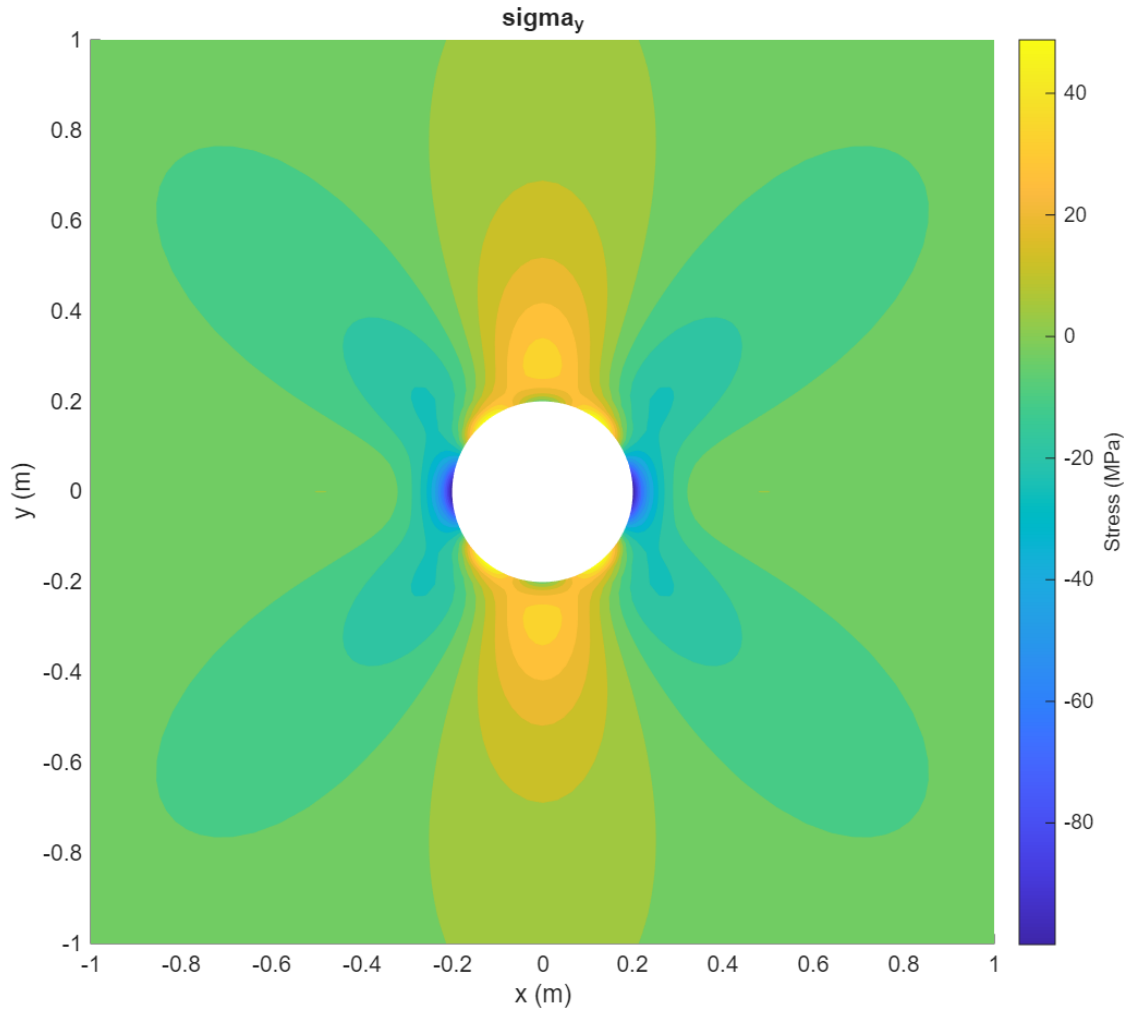
$$\sigma_{VM} = \sqrt{\sigma_x^2 + \sigma_y^2 - \sigma_x\sigma_y + 3\tau_{xy}^2}$$

- Stress concentration factor:

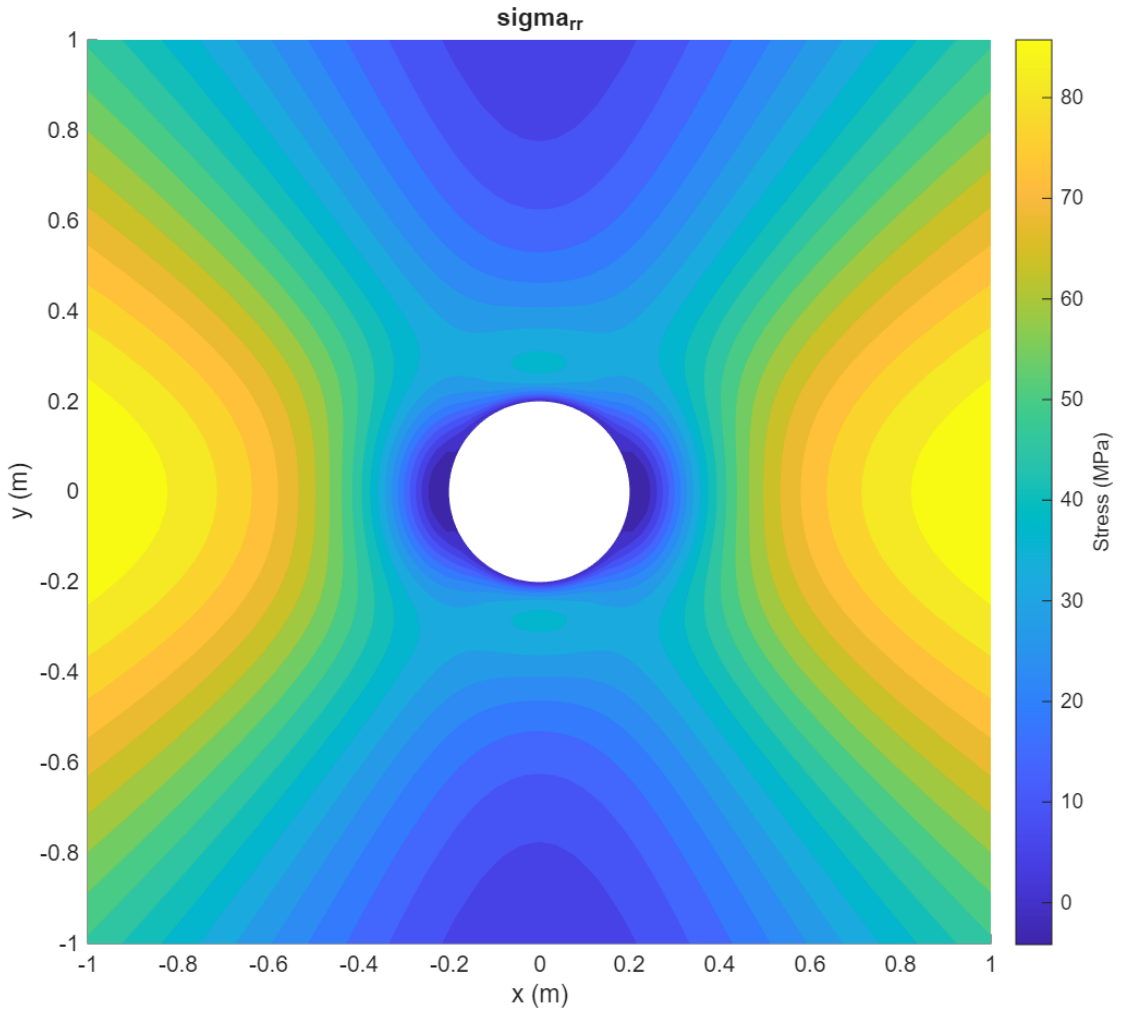
$$K = \frac{\sigma_{max}}{\sigma_{ave_{max}}} = \frac{\max(\sigma_x)}{\sigma_0}$$

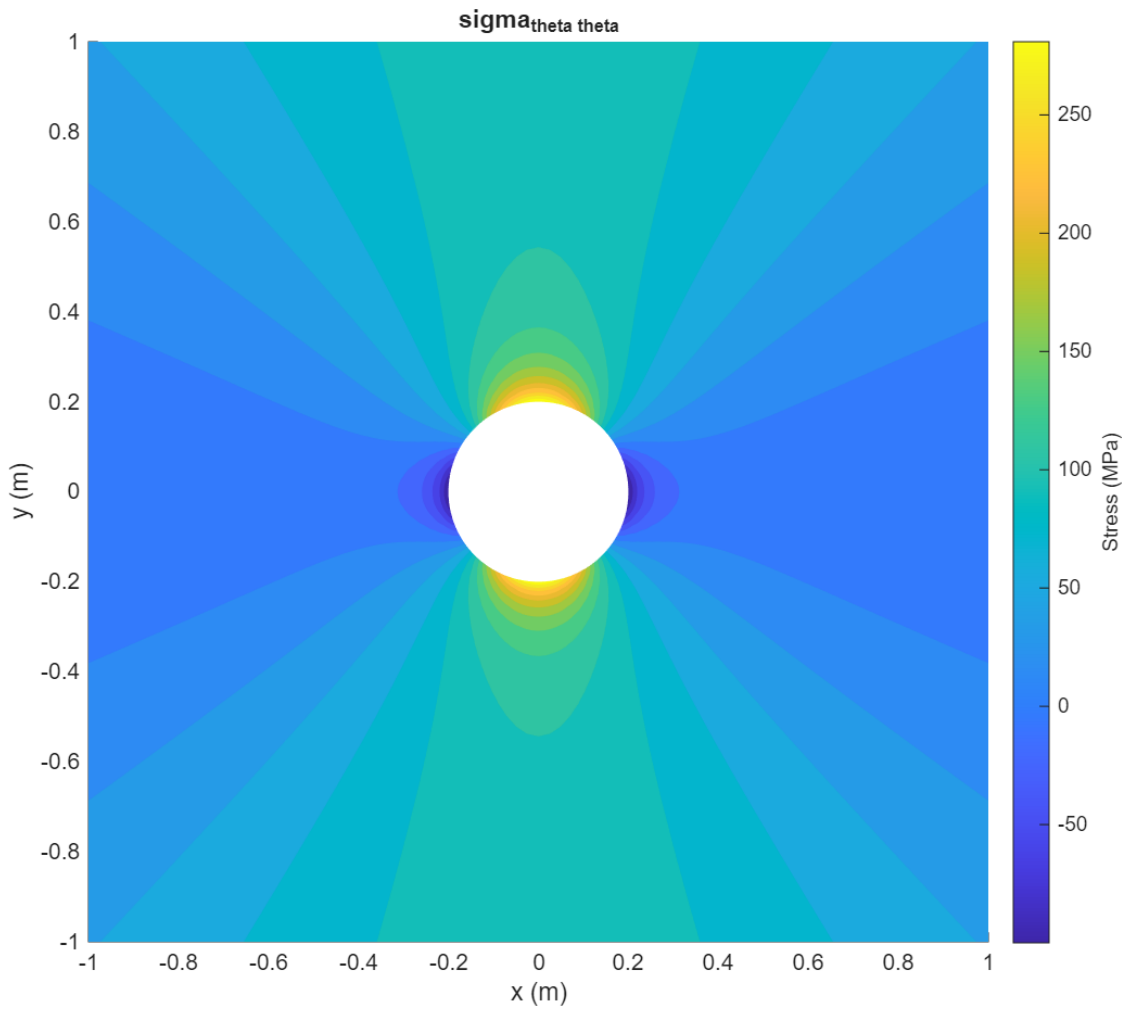
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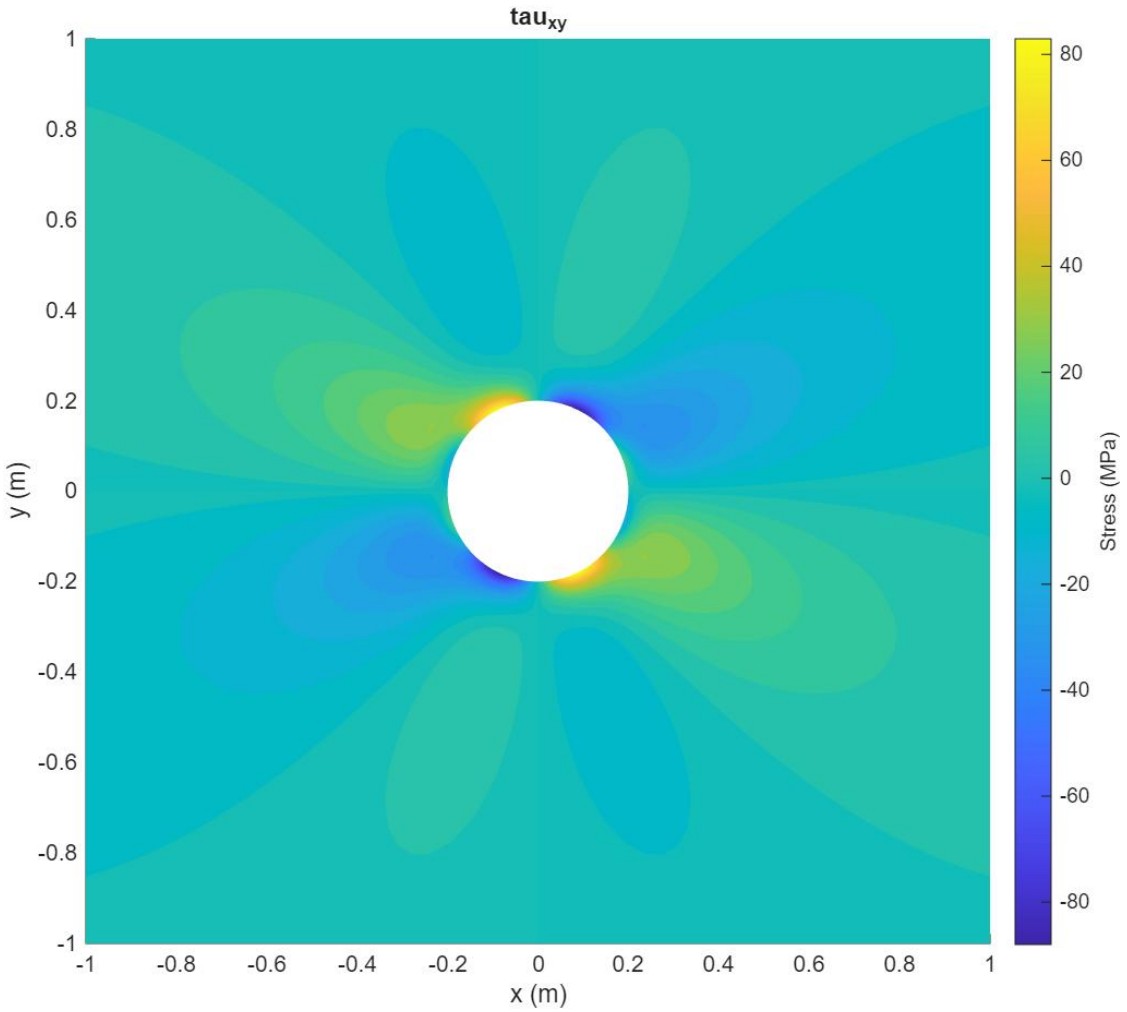


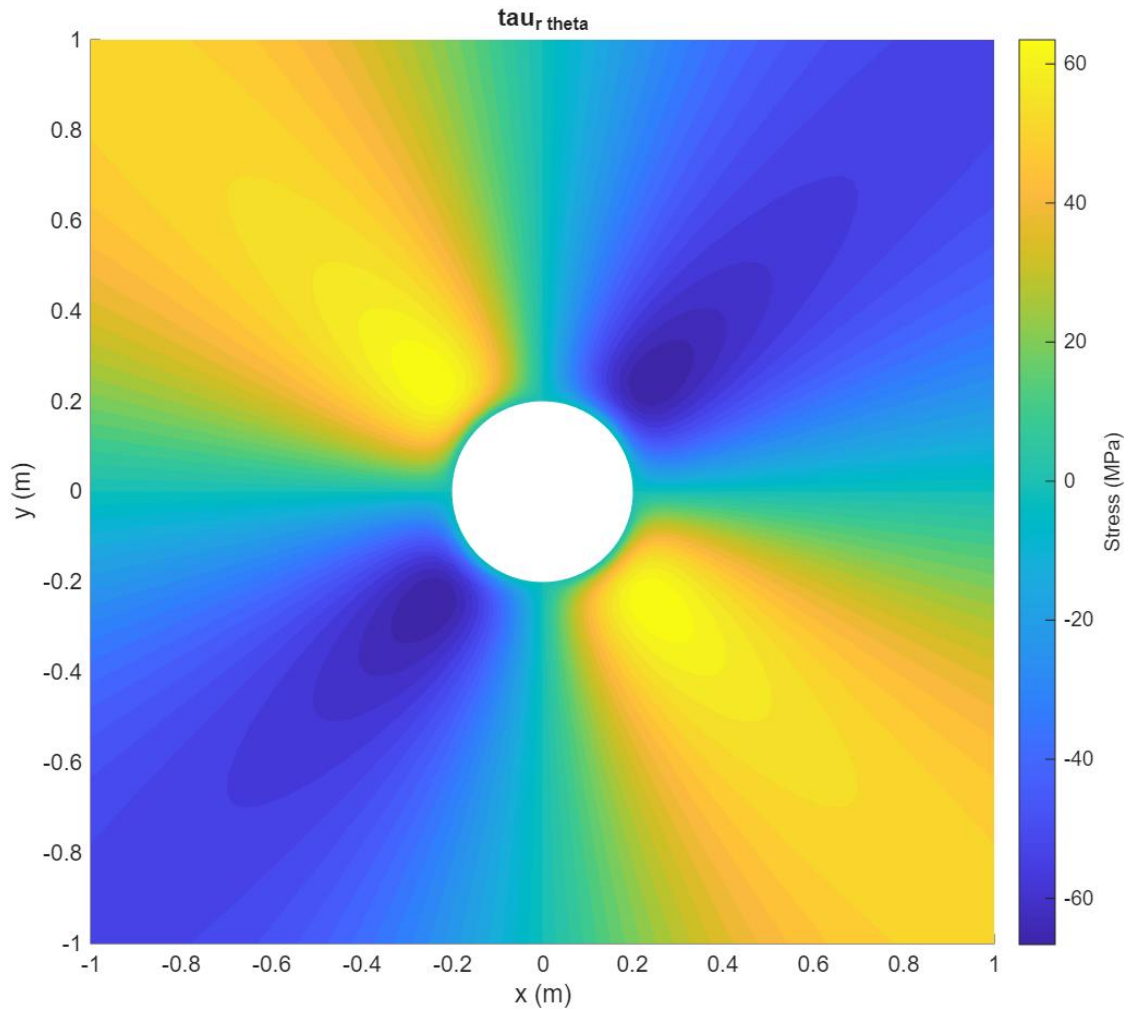
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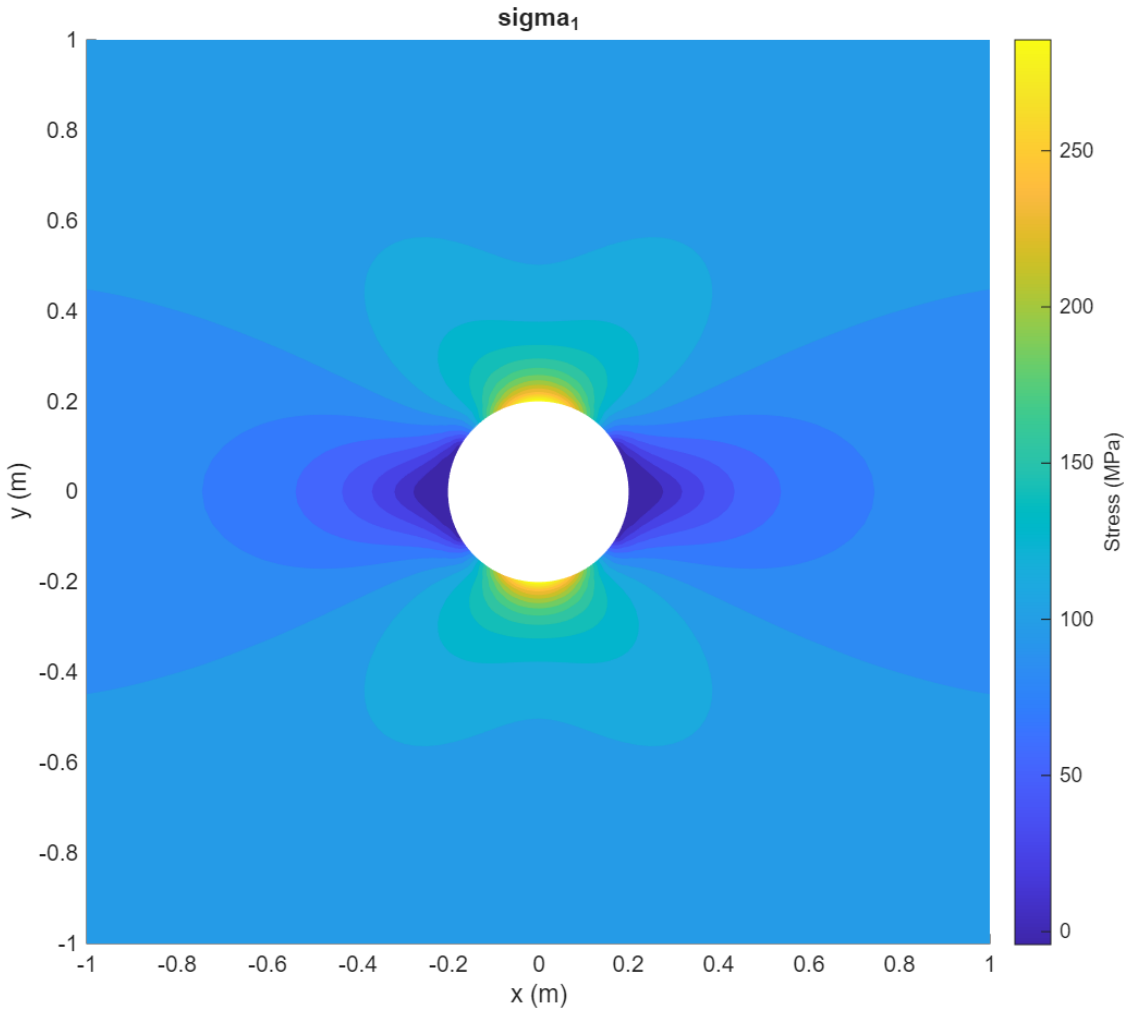


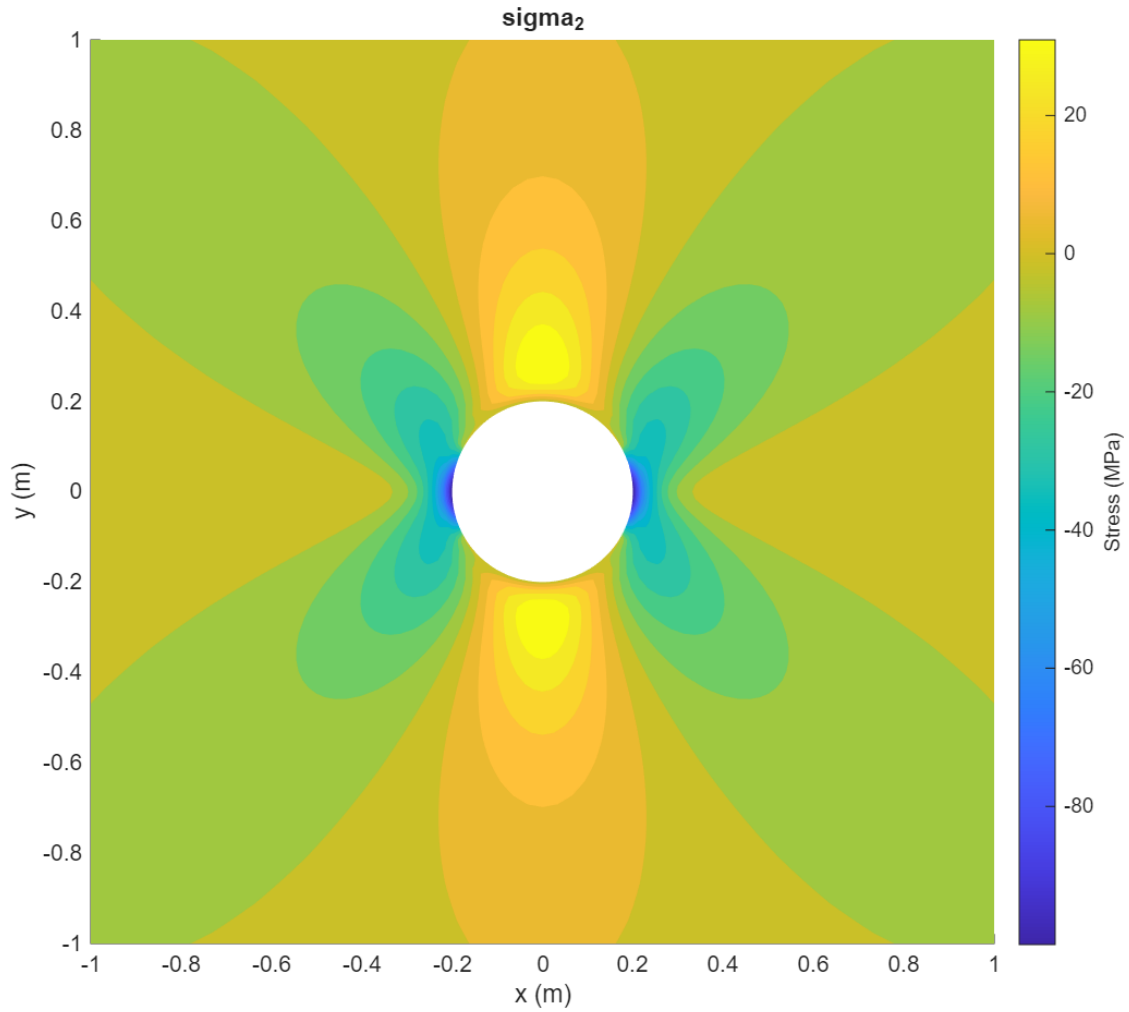
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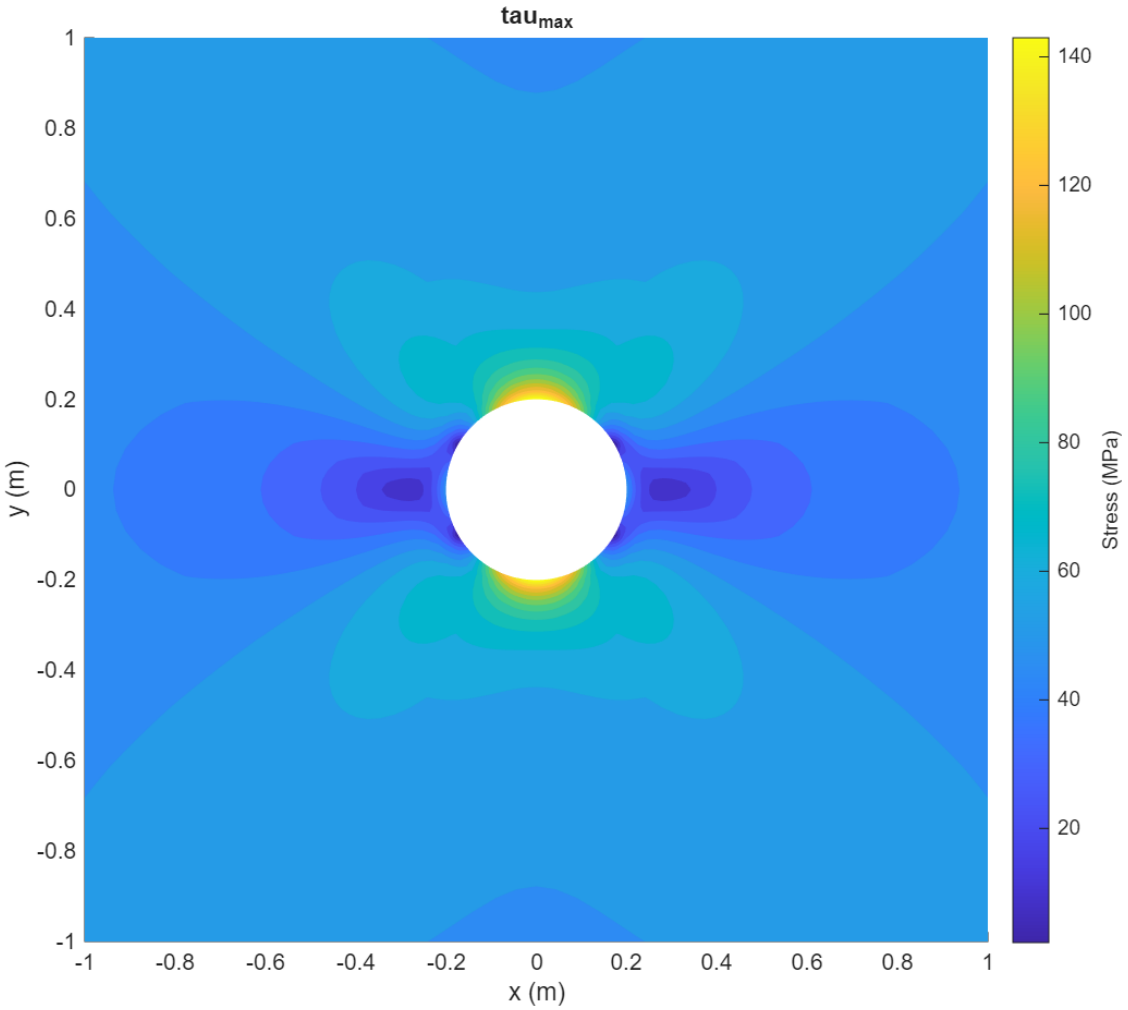


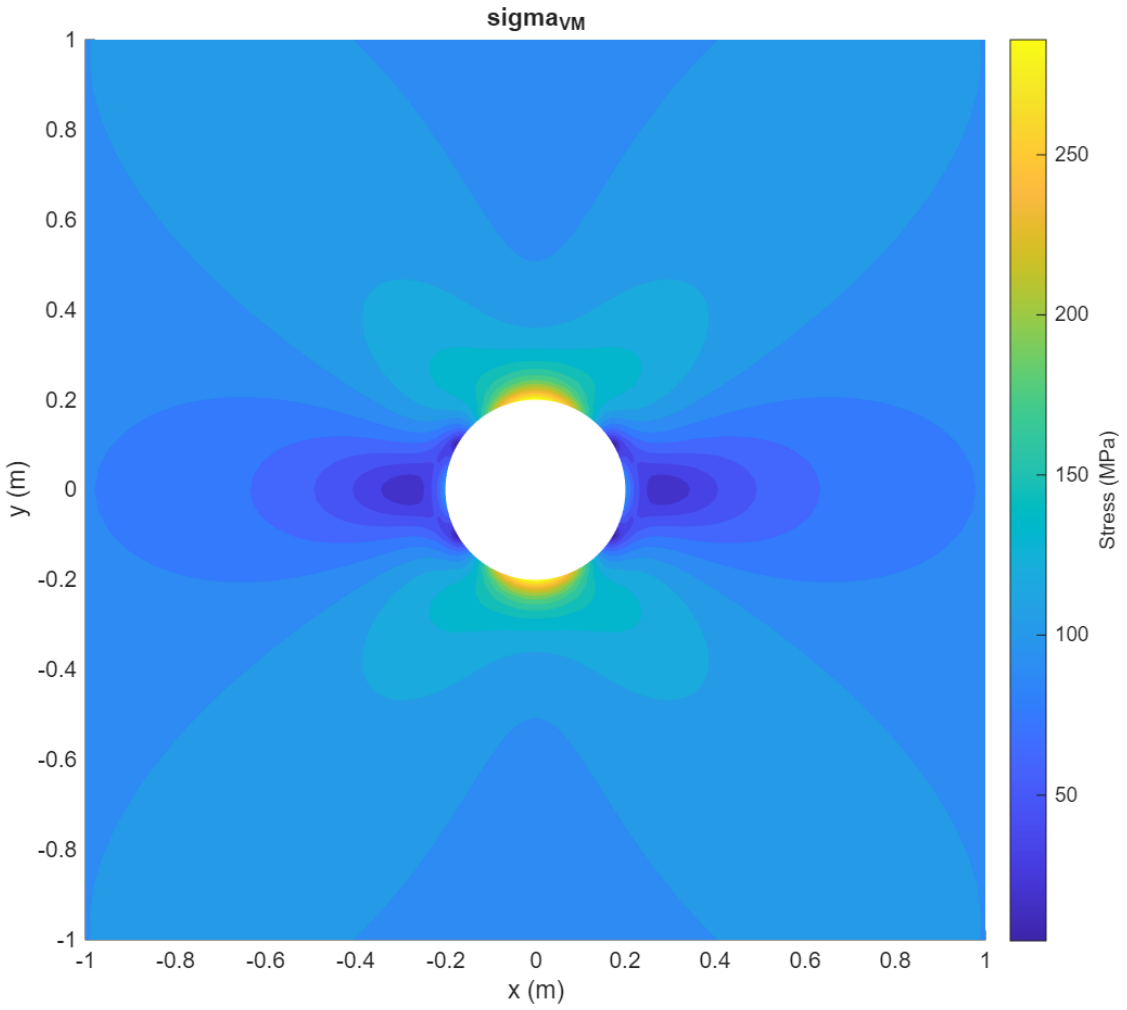
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>> stresscontour
Stress concentration factor: 3.00
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Some of the boundary conditions at the sides of the plate aren't met (zero traction surface) which is weird, but probably due to the stress distribution being unable to fully model boundaries. [Time spent: 1.5 hrs]